

BEHAVIORAL FINANCE (MBA680) | GROUP PROJECT

Personal Investment Advisor and Robo-Advisory Project

A step by step walk-through of Steps 1 to 6, with the survey questions used at each step

Portal: siddharthjaiswal05.github.io/mba_680_assignment_2

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The Six Steps at a Glance

The map. Each of the next nine slides takes one step and shows the questions, the method and the result together.

Every step, the part of the form it used, and what it produced

Step	What it required	Part of the form used	How it was done	What came out
1	Pick a peer group and build the survey	Section 1, About you	Eight profile questions, roll number instead of a name	Four anonymous investor profiles
2	Measure the attitude to risk	Section 2, eight ladder rows	A sure amount against a double or nothing coin flip	A risk pattern for each person
3	Measure the attitude to waiting	Section 3, six now or later rows	Rs 10,000 today against a larger amount later	A yearly discount rate for each person
4	Fit the curves and score the personality	Sections 4, 5 and 6	Probability equivalence, then a least squares fit; BFI-2 scored by facet	Risk aversion A, curvatures alpha and beta, five domain scores
5	Build the portfolio	Nothing new, it reuses Steps 2 to 4	Maximise $E(r)$ minus half A sigma squared along the capital allocation line	An advised equity weight for each person
6	Write the advice and ship the tool	Section 4 scores plus Section 7	Rules that fire on the respondent's own answers	Personal commentary and a live portal

The seven Section 7 questions are our own addition to the class form. They are introduced on the Step 5 behavioral slide, where they are used.

Choose the Peer Group and Build the Survey

Requirement: field an investor survey on a peer group and record it in a form that an estimator can read.

The eight questions in Section 1, as they appeared

Question on the form	Answer type
Roll number	Text, used instead of a name
Age	Years
Money coming in each month	One of five bands
Money you can invest (Rs)	Rupees
How much investing have you done?	None to Substantial
How many years will you keep it invested?	Years
What is the money mainly for?	One of four goals
Anything that limits you?	Optional free text

How it was done

Fielded to four classmates through the portal. The export of their answers is the file the estimator reads.

What came back

Roll	Investable wealth	Experience	Horizon	Primary goal
251250039	Rs 90,000	Moderate	5 yrs	Wealth creation
251250047	Rs 2 lakh	None	10 yrs	Wealth creation
251250002	Rs 10 lakh	Limited	6 yrs	Wealth creation
251250030	Rs 1 lakh	Limited	15 yrs	Specified goal

How the step was completed

Four responses collected and transcribed into one machine readable file.

Output of Step 1

engine/responses_real.json, the single input to every later step.

Measure the Attitude to Risk

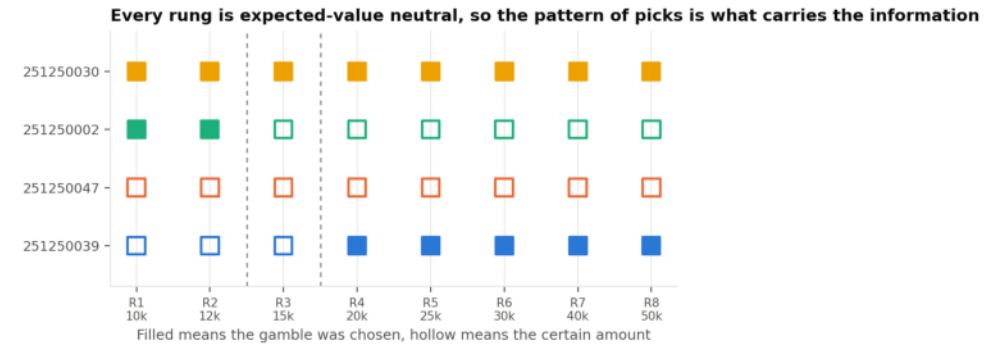
Requirement: put a sure amount against a fair gamble and see which one the respondent takes, row by row.

Section 2, all eight rows exactly as asked

#	A: you get for sure	B: coin flip
1	Rs 10,000	Rs 20,000 or nothing, 50:50
2	Rs 12,000	Rs 24,000 or nothing, 50:50
3	Rs 15,000	Rs 30,000 or nothing, 50:50
4	Rs 20,000	Rs 40,000 or nothing, 50:50
5	Rs 25,000	Rs 50,000 or nothing, 50:50
6	Rs 30,000	Rs 60,000 or nothing, 50:50
7	Rs 40,000	Rs 80,000 or nothing, 50:50
8	Rs 50,000	Rs 1 lakh or nothing, 50:50

How it was done

The sure amount equals the gamble's average on every row, so one utility curve means the same side all eight times. We read the pattern of the picks.



How the step was completed

All four patterns were classified. 251250047 took the sure amount every time and 251250030 took the coin flip every time.

What it told us

251250039 and 251250002 switch part way down, at rows 4 and 3, exactly as prospect theory predicts.

Measure the Attitude to Waiting

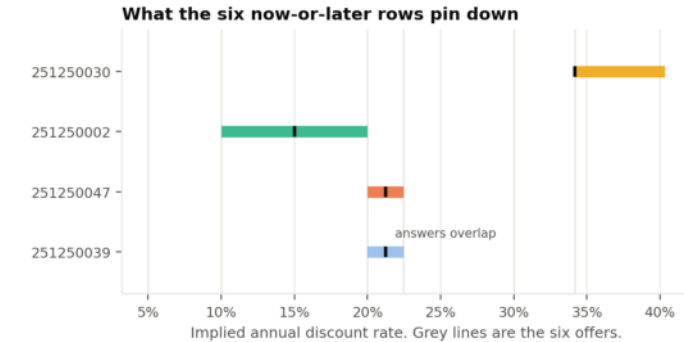
Requirement: offer money now against more money later and find the return that makes waiting worthwhile.

Section 3, all six rows exactly as asked

#	Take now	Or wait and get	Wait how long
1	Rs 10,000	Rs 10,500	1 year
2	Rs 10,000	Rs 11,000	1 year
3	Rs 10,000	Rs 12,000	1 year
4	Rs 10,000	Rs 13,000	1 year
5	Rs 10,000	Rs 15,000	2 years
6	Rs 10,000	Rs 18,000	2 years

How it was done

Each row implies a yearly return. The lowest offer waited for and the highest one turned down bracket the personal discount rate, and the midpoint of that bracket is the point estimate.



How the step was completed

A discount rate was produced for all four. 251250039 came out near 21.2% a year and 251250002 near 15.0%.

What it told us

Every rate sits above any return on offer, and 251250030 preferred cash at every rung. Worth a conversation about the real horizon.

Fit the Utility Curve

Requirement: elicit a utility curve over wealth by the probability equivalence method from the lecture.

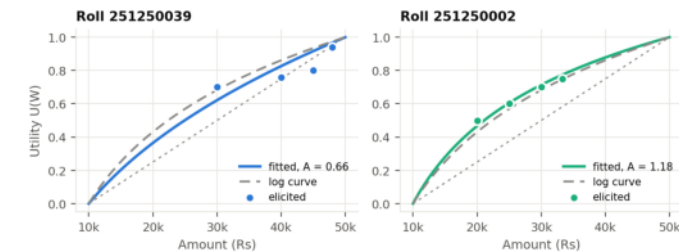
Section 5, with one respondent's own answers alongside

Item	Question on the form	251250039 said
A	The lowest amount that feels relevant to you	Rs 10,000
B	The highest amount that feels relevant to you	Rs 50,000
U1	Rs 30,000 for sure, or play the game. What winning chance makes the two feel the same?	70 in 100
U2	Rs 40,000 for sure, or play the game. What winning chance makes the two feel the same?	76 in 100
U3	Rs 45,000 for sure, or play the game. What winning chance makes the two feel the same?	80 in 100
U4	Rs 47,999 for sure, or play the game. What winning chance makes the two feel the same?	94 in 100

How it was done

Set $U(A) = 0$ and $U(B) = 1$. At indifference the stated chance is the utility of the sure amount, so $U(X) = p$. Those points were fitted with a constant relative risk aversion curve by least squares.

Elicited utility curves, with the log curve the lecture suggests



The fitted curves

Roll	Range A to B	Risk aversion	Fit R2
251250039	Rs 10,000 to Rs 50,000	0.66	0.202
251250047	To come	To come	To come
251250002	Rs 10,000 to Rs 50,000	1.18	0.948
251250030	To come	To come	To come

Both curves came out concave, so both respondents are risk averse: $A = 0.66$ and $A = 1.18$. That coefficient is the A used directly in Step 5.

Fit the Value Function Over Gains and Losses

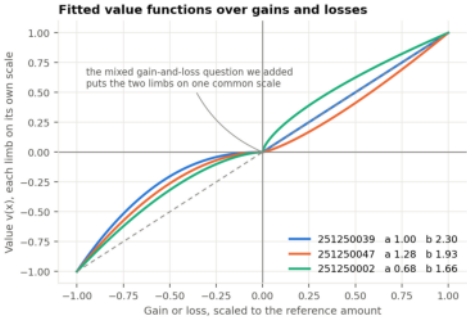
Requirement: measure how gains and losses are coded against a reference point, as prospect theory sets it out.

Section 6, all seven fields exactly as asked

Item	Question on the form	Answer scale
W0	The amount you measure gains and losses against	Rupees
V1	Win Rs 10,000 for sure, or play for Rs 20,000	Winning chance
V2	Win Rs 20,000 for sure, or play for Rs 40,000	Winning chance
V3	Win Rs 30,000 for sure, or play for Rs 60,000	Winning chance
V4	Lose Rs 10,000 for sure, or risk losing Rs 20,000	Chance of the bigger loss
V5	Lose Rs 20,000 for sure, or risk losing Rs 40,000	Chance of the bigger loss
V6	Lose Rs 30,000 for sure, or risk losing Rs 60,000	Chance of the bigger loss

How it was done

The gamble pays double or nothing, so $v(X) = p$ times $v(2X)$ at indifference. With $v(x) = x$ to the power alpha the sure amount cancels and alpha is minus log base 2 of p.



The fitted parameters

Roll	Gain alpha	Loss beta	Spread across rows
251250039	1.00	2.30	0.81 and 1.10
251250047	1.28	1.93	0.51 and 0.32
251250002	0.68	1.66	0.49 and 1.64
251250030	To come	To come	To come

All three gain rows measure the same curvature, so the spread between them is a direct read on how reliably the answer was given.

Score the Personality Profile

Requirement: collect a standard personality measure so the advice can speak to behaviour, not only to risk.

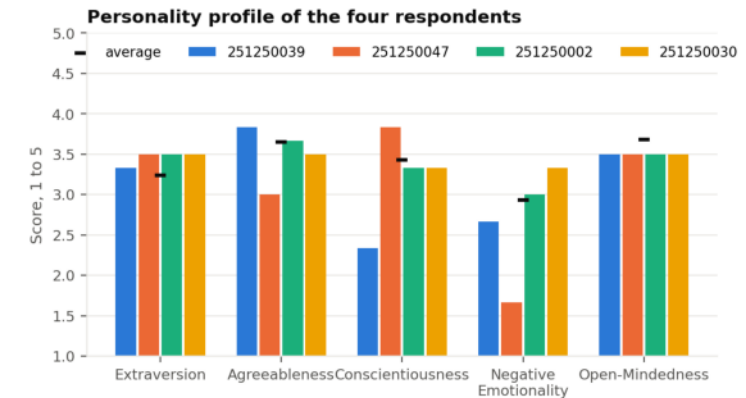
Section 4, thirty short items, one sample from each domain

Domain	A sample item from the thirty	Scale
Extraversion	Is outgoing, sociable.	1 to 5
Agreeableness	Is compassionate, has a soft heart.	1 to 5
Conscientiousness	Is systematic, likes to keep things in order.	1 to 5
Negative emotionality	Can be tense.	1 to 5
Open-mindedness	Is curious about many different things.	1 to 5

How it was done

Thirty items from the BFI-2, two per facet, chosen so all fifteen facets are covered within the thirty question cap. Reverse keyed items were flipped, facets averaged, domains averaged from their facets, then placed against the published norming means.

1 means not like me at all and 5 means very much like me.



How the step was completed

All four answered all thirty items, so every domain and facet score is complete.

Output of this step

Five domain scores and fifteen facet scores per person.

Build the Mean-Variance Portfolio

Requirement: combine the measured risk aversion with the Problem Statement 1 portfolio and set the equity weight.

The inputs, all carried forward rather than asked again

Input	Value used	Where it came from
Risky portfolio	24.9% return, 18.0% risk	Problem Statement 1
Risk-free rate	6.5%	Treasury bill yield
Risk aversion A	0.66 and 1.18	The Step 4 utility curve
Borrowing	Capped at zero	Student investor constraint

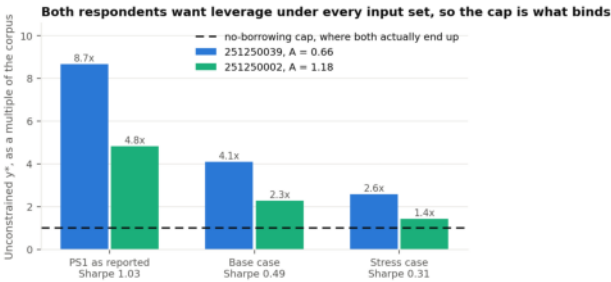
How it was done

Maximise $U = E(r)$ minus half A sigma squared along the capital allocation line. The optimum is $y^* = [E(r_P) \text{ minus } r_f] \text{ divided by } A \text{ times sigma}_P \text{ squared}$, capped at one because borrowing is not available.

Three return scenarios are carried through, because the expected return is the least certain input in the whole exercise.

The recommendation, base case inputs

Roll	A	y* uncapped	Advised equity	Return	Risk
251250039	0.66	4.11x	100%	15.2%	18.0%
251250047	To come	To come	To come	To come	To come
251250002	1.18	2.29x	100%	15.2%	18.0%
251250030	To come	To come	To come	To come	To come



Both respondents with a fitted curve want more than their whole corpus in the risky portfolio, so the cap sets the answer: hold 100%, which is Rs 90,000 and Rs 10 lakh.

Build the Behavioral Portfolio

Requirement: go past mean-variance and size the layered portfolio that behavioral portfolio theory describes.

Section 7, the questions we added to the class form

Item	Question added to the form	What it measures
E1	A coin flip. Tails you lose Rs 10,000, heads you win G. How big must G be?	Loss aversion, lambda
E2	A 10 in 100 chance of Rs 10,000. What sure amount feels as good? Same at 90 in 100.	Probability weighting
E3	What is the smallest amount you must still have at the end?	The safety floor
E4	How often could you live with falling below that amount?	Shortfall tolerance
E5	What amount at the end would you call a success?	The aspiration level
E6	Split 100 points between staying safe and a real shot at the big number.	Security and potential split
E7	What would you give up to raise your chance of success by 10 points?	Weight on the aspiration

How it was done

Each layer needs one number the class form does not collect, so we wrote the seven shortest questions that supply them. They take about a minute to answer.

What the layered portfolio needs

A floor in rupees, a tolerance for breaching it, an aspiration level, and loss aversion. Together they split the corpus into a safety sleeve and a potential sleeve, each priced off the same frontier.

How it was completed

The engine, the estimators and the layer solver are built and tested, and all seven questions are live in the portal. The code path runs end to end on worked inputs.

The next pass

The four respondents answer the seven added questions, and the same run then prints a sized behavioral portfolio beside the mean-variance one.

Turn the Numbers Into Advice

Requirement: write the commentary and the behavioral guidance that goes back to each respondent.

The rules, and the answer that triggers each one

Guidance	Fires when	Fired for
Automate the monthly contribution	Conscientiousness below the group mean	251250039
Fix the review dates in advance	Organization facet below 3 out of 5	251250039
Step the equity weight up over time	Low negative emotionality and a long horizon	251250047
Show the downside in rupees, not percent	A switch part way down the risk ladder	251250002
Confirm the real horizon first	Cash preferred at every rate offered	251250030

251250039: build in the follow-through

Conscientiousness 2.33. Automate the contribution and rebalance on a threshold.

251250047: methodical and calm

Conscientiousness 3.83 and the group's lowest negative emotionality at 1.67. A glide path suits this well.

251250002: frame it in rupees

Balanced across every domain. The switch at row 3 suggests risk is judged in nominal amounts, so show the downside as a rupee figure.

251250030: start with a conversation

Took the gamble throughout and preferred cash now at every rate. A short interview will establish the real horizon before the plan is set.

How it was done

Risk aversion is already measured in Step 4, so personality is used only where it predicts something else: whether the client stays with the plan. Each rule names the answer that set it off, so nothing in the advice is unexplained.

How Each Step Was Completed

The same six steps again, this time with the evidence for each one.

Step by step, what exists at the end of the project

Step	How it was completed	Where to see it
1. Peer group and survey	Eight profile questions fielded to four classmates, answers stored under roll numbers and exported to one file	Portal Section 1, responses_real.json
2. Risk attitude	All eight ladder rows answered by all four; each pattern classified and the expected utility check reported	Report Section 2, figure c3
3. Time preference	All six rows answered by all four; a discount rate bracket and a midpoint estimate produced for each	Report Section 3, figure c4
4. Curves and personality	Utility curves fitted for two respondents, value curves for three, and all thirty personality items scored for all four	Report Section 4, figures c1, c2, c5
5. Portfolio	Advised equity weight computed for every respondent with a fitted curve, under three return scenarios	Report Section 5, figures c6, c7
6. Advice and tool	Written commentary per respondent, plus a live portal that fields the form and repeats the whole calculation in the browser	Report Section 6, the portal
Beyond the brief	Seven extra questions added to the form so the behavioral layers can be sized from stated preferences	Portal Section 7

Two respondents are scheduled for a short re-elicitation so their utility and value curves join the Step 5 run.

The Robo-Advisory Portal

One static page, no server and no sign-in, so it opens on a phone in a classroom.

Fields the whole form

All six class sections plus the seven added questions, written in plain English, with autosave and a one-click export of each respondent's answers.

Fits and plots live

Draws the respondent's own utility curve, value function and weighting function in the browser, using the same estimators that produced this deck.

Builds both portfolios

The mean-variance allocation and the behavioral layers, against the real Problem Statement 1 frontier, with commentary generated from the respondent's own answers.

siddharthjaiswal05.github.io/mba_680_assignment_2

What We Established, and What Comes Next

Sample

Four respondents, so the conclusions are about the method rather than a population.

Return input

The least certain input, which is why three scenarios are carried through.

Horizon

Single-period models, before contributions, taxes and costs.

Distribution

Normality assumed, so tail figures are the optimistic end.

Universe

Current index membership, inherited from Problem Statement 1.

- Every one of the six steps is traceable to the exact question that produced it, and every number in this deck is computed from a submitted answer.
- Risk aversion is measured from the utility curve, and the ladder is read as the pattern it genuinely reveals.
- Seven added questions complete the behavioral portfolio, and they are already live in the portal.
- The next pass collects those answers and re-elicits two respondents, which closes out Steps 4 and 5 for the full group.

Measure what the data supports, and extend the instrument where it does not.